

Title: Catering Management System

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INTRODUCTION

Background

The catering industry in Kenya plays an important role in supporting events such as weddings, corporate meetings, conferences, funerals and social gatherings. However, most small and medium-sized catering businesses largely rely on informal tools such as handwritten records, spreadsheets, and phone calls to manage their operations.

Key operational activities such as booking management, client quotations, invoicing, menu costing, inventory tracking, equipment management, and staff scheduling are not handled efficiently which often introduce significant inefficiencies as the business grows and scales.

Problem Statement

There is lack of an efficient digital system that automates internal catering business processes - preventing catering companies from managing bookings, quotations, inventory, staff and customer records efficiently within a single platform.

As a result there is risk of:

Operational Inefficiency

Manual record keeping leads to booking conflicts, miscommunication

Financial Leakage

Inaccurate quotations, unpaid invoices, and untracked expenses result in revenue loss and poor financial visibility.

Poor Customer Experience

Lack of customer history and inconsistent service delivery reduces repeat business and damages reputation

Proposed Solution

The proposed system is a web-based Catering Operations Management System designed to digitize and automate the internal operational workflows of catering businesses.

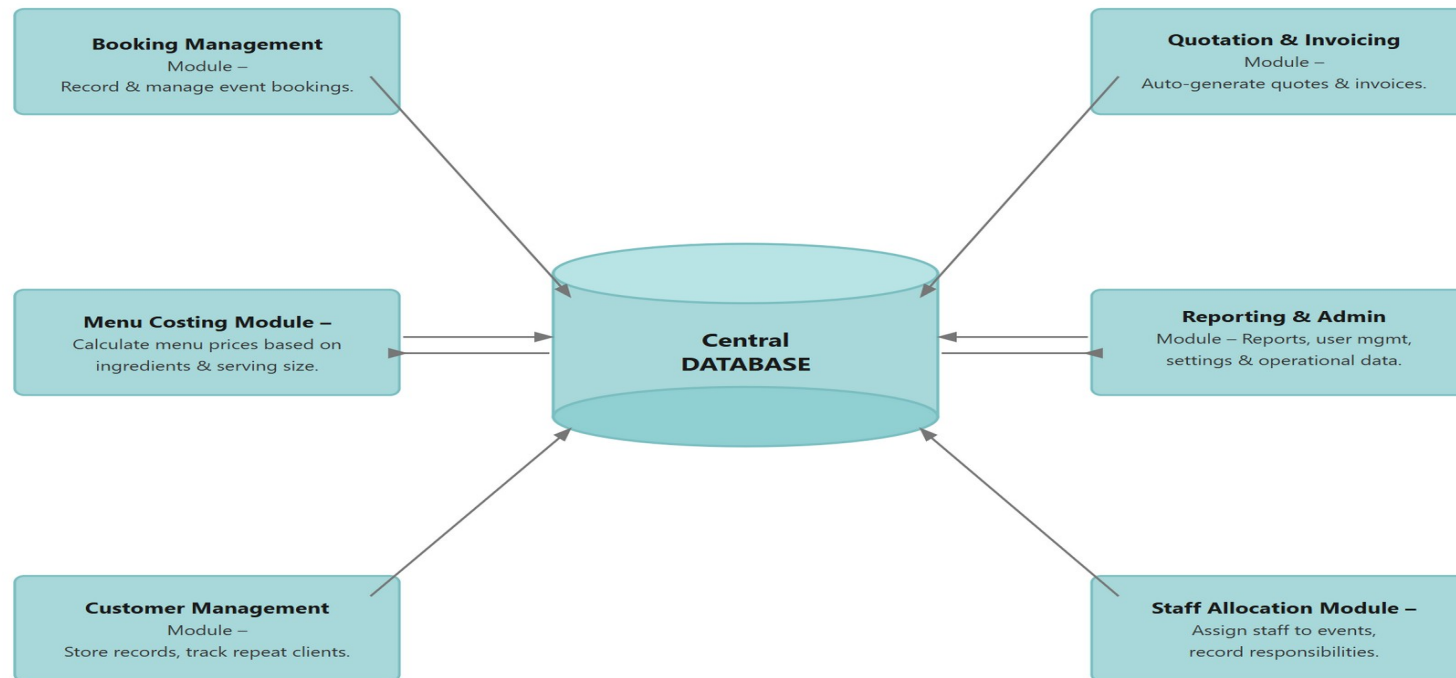
The system will allow catering companies to record client bookings, generate quotations and invoices automatically, manage menu pricing, track inventory and equipment, assign staff to events, and maintain customer records.

By centralizing operational data in a single system, catering businesses will be able to improve organization, reduce manual errors, and enhance overall service efficiency.

Objectives

1. To develop a web-based Catering Management system that automates booking, quotation, and operational processes.
2. To design a Booking Management module that records and manages customer catering event requests.
3. To implement automated Quotation & Invoice generation based on menu selection and event size.
4. To develop an Inventory & Equipment Tracking module for managing catering resources efficiently.
5. To implement a reporting module that generates operational and financial reports for management.

Conceptual Diagram



Tools to be Used

1. Frontend: React
2. Backend: Springboot (Java)
3. Database: PostgreSQL
4. Version Control: Github
5. Design: Figma

System Modules & Functionality

01 Booking Management

Client Books an Event

Event details (date, location, guest count, menu) are recorded. The system checks for scheduling conflicts and confirms the booking.

02 Menu Costing

Menu Costing Calculated

This Module calculates the total service cost based on menu items, portion sizes, and selected service packages.

03 Quotation & Invoice

Quotation & Invoice Generated

A Quotation is sent to the client. Once approved, an invoice is generated and payment status is tracked automatically.

04 Staff Management & Inventory Module

Staff & Inventory Assigned

Staff are assigned roles for the event. The inventory module checks equipment and flags any resource shortfalls.

05 Reporting & Admin Module

Reports & Admin Review

The Admin module compiles data from all modules – bookings, payments, inventory, staff – into operational and financial reports.